Probabilistic Modeling of Data Labeling Reliability

Probability Theory × Machine Learning × Crowdsourcing

Lecture Notes — October 2026

1. Problem Formulation

Suppose we have a training dataset of n examples, $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^{n}$, where y_j is the unknown ground-truth label. There are k available annotators. How many annotations per example are needed so that the final aggregated label is correct with probability at least $m \in [1/2, 1)$?

We begin with binary classification, a unique true label, and annotator errors that are independent conditional on that label. Importantly, independence alone is insufficient: we also need information about individual annotator reliability.

2. Homogeneous Annotators: Bernoulli and Binomial Models

Suppose each annotator independently labels an example correctly with probability $q > 1/2$. Define

$$X_i = \begin{cases} 1 & \text{annotator } i \text{ is correct,} \\ 0 & \text{otherwise.} \end{cases} \qquad X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(q).$$

For r annotators, $S_r = \sum_{i=1}^{r} X_i \sim \text{Binomial}(r, q)$.

2.1 Majority voting

Take $r = 2t + 1$ to avoid ties. With binary labels, majority voting is correct precisely when $S_r \geq t + 1$, hence

$$\boxed{A_r(q) = \mathbb{P}(S_r \geq t + 1) = \sum_{j=t+1}^{2t+1} \binom{2t+1}{j} q^j (1-q)^{2t+1-j}.}$$

For target reliability m, the minimum odd number of annotators is

$$\boxed{r_{\min}(q, m) = \min\{r \in \{1, 3, 5, \ldots\} : A_r(q) \geq m\}.}$$

If the pool contains only k annotators, feasibility also requires $r_{\min} \leq k$.

2.2 Reliability amplification

If $q > 1/2$, then $S_r/r \xrightarrow{P} q$ by the law of large numbers, so

$$\lim_{\substack{r \to \infty \\ r \text{ odd}}} A_r(q) = 1.$$

This is closely connected to the *Condorcet Jury Theorem*. Independence and individual accuracy above chance are key assumptions.

3. Heterogeneous Annotators: Poisson Binomial Distribution

Now annotator i has error probability p_i and accuracy $q_i = 1 - p_i$. The indicator $X_i \sim \text{Bernoulli}(q_i)$, but the X_i are not identically distributed. If they remain independent,

$$\boxed{S_r = \sum_{i=1}^{r} X_i \sim \text{PoissonBinomial}(q_1, \ldots, q_r).}$$

For $s \in \{0, \ldots, r\}$,

$$\mathbb{P}(S_r = s) = \sum_{\substack{A \subseteq \{1,\ldots,r\} \\ |A|=s}} \prod_{i \in A} q_i \prod_{j \notin A} (1 - q_j).$$

For odd $r = 2t + 1$, ordinary majority-vote reliability remains $A_r(\mathbf{q}) = \mathbb{P}(S_r \geq t + 1)$.

3.1 Counterexample: more voters can hurt

Take $(q_1, q_2, q_3) = (0.9, 0.8, 0.6)$. At least two must be correct:

$$\begin{aligned} A_3 = {} & (0.9)(0.8)(0.6) + (0.9)(0.8)(0.4) \\ & + (0.9)(0.2)(0.6) + (0.1)(0.8)(0.6) = 0.876. \end{aligned}$$

Yet the strongest annotator alone has reliability 0.9. Thus unweighted majority voting can be worse than using a single reliable annotator.

3.2 Dynamic programming for exact probabilities

Let $D_i(s) = \mathbb{P}(\sum_{j=1}^{i} X_j = s)$, with $D_0(0) = 1$. Conditioning on X_i gives

$$\boxed{D_i(s) = (1 - q_i) D_{i-1}(s) + q_i D_{i-1}(s - 1).}$$

Summing $D_r(s)$ over the majority tail computes the exact reliability in $O(r^2)$ time and $O(r)$ space. Out-of-range states are interpreted as zero.

4. Bayesian Weighted Majority Voting

Let the true binary label be $Y \in \{0, 1\}$ and annotator i report $Z_i \in \{0, 1\}$. Assume *symmetric* conditional error probability $\mathbb{P}(Z_i \neq Y \mid Y) = p_i$ and conditional independence given Y. Set $\mathbb{P}(Y = 1) = \pi$.

Bayes' theorem gives posterior log-odds:

$$\boxed{\log \frac{\mathbb{P}(Y = 1 \mid Z_1, \ldots, Z_r)}{\mathbb{P}(Y = 0 \mid Z_1, \ldots, Z_r)} = \log \frac{\pi}{1 - \pi} + \sum_{i=1}^{r} \log \frac{1 - p_i}{p_i} (2Z_i - 1).}$$

Define $w_i = \log \frac{1-p_i}{p_i}$. The Bayes-optimal rule under zero–one loss predicts label 1 when

$$\boxed{\log \frac{\pi}{1 - \pi} + \sum_{i=1}^{r} w_i (2Z_i - 1) > 0.}$$

For balanced prior $\pi = 1/2$, the prior term vanishes. If $p_i < 1/2$, the weight is positive; if $p_i = 1/2$, it is zero; if $p_i > 1/2$, it is negative. This formula is specific to binary symmetric errors; class-dependent and multiclass errors need a confusion-matrix likelihood.

5. Concentration Inequalities and Sample Complexity

For identical annotators with $q > 1/2$, Hoeffding's inequality implies

$$\mathbb{P}\left(\frac{S_r}{r} \le \frac{1}{2}\right) \le \exp\left(-2r(q-\tfrac{1}{2})^2\right).$$

For odd r, majority error is included in this event. Therefore a *sufficient*, usually conservative, condition for correctness probability at least m is

$$\boxed{r \ge \frac{\ln(1/(1-m))}{2(q-\frac{1}{2})^2}.}$$

Round upward to an allowable odd count. For independent, heterogeneous Bernoulli indicators, writing $\bar{q} = r^{-1}\sum_i q_i > 1/2$ yields

$$\boxed{\mathbb{P}(\text{majority error}) \le \exp\left(-2r(\bar{q}-\tfrac{1}{2})^2\right).}$$

This concerns *unweighted* majority voting, not Bayesian weighted voting.

6. Per-Example versus Whole-Dataset Reliability

6.1 Per-example guarantee

The objective $\mathbb{P}(\hat{Y}_j = Y_j) \ge m$ for every example j does not directly depend on the dataset size n.

6.2 All labels simultaneously correct

A stronger objective is $\mathbb{P}(\hat{Y}_1 = Y_1, \ldots, \hat{Y}_n = Y_n) \ge m$. If final correctness events are independent and all have reliability A_r, then $(A_r)^n \ge m$, or

$$\boxed{A_r \ge m^{1/n}.}$$

Without independence across examples, the union bound implies

$$\mathbb{P}(\text{at least one error}) \le \sum_{j=1}^{n} \mathbb{P}(\hat{Y}_j \ne Y_j).$$

Thus requiring $\mathbb{P}(\hat{Y}_j \ne Y_j) \le (1-m)/n$ for all j suffices to ensure a whole-dataset guarantee of at least m.

7. Beyond the Basic Model: Real-World Labeling

7.1 Correlated errors

Annotators may make the same mistake because an image is ambiguous, blurred, or misleading. Then $\mathbb{P}(E_i \cap E_j) \neq \mathbb{P}(E_i)\mathbb{P}(E_j)$. Positive error correlations can greatly diminish reliability amplification.

7.2 Item-dependent difficulty

A useful extension sets $p_{ij} = \mathbb{P}(\text{annotator } i \text{ errs on example } j)$. Accuracy can vary with blur, glare, occlusion, camera rotation, and rolling digits.

7.3 Multiclass labels and Dawid–Skene

For C classes, annotator i can be described by a confusion matrix $\Pi_i(a, b) = \mathbb{P}(Z_i = b \mid Y = a)$. Dawid–Skene-type models jointly infer latent true labels and annotator confusion matrices, subject to identification assumptions.

7.4 Continuous geometric labels

An oriented bounding box is $b = (c_x, c_y, w, h, \theta)$; two boxes can disagree slightly without either being plainly wrong. One possible binary quality indicator is

$$X_i = \mathbf{1}\big\{\,\mathrm{RotatedIoU}(\hat{b}_i, b^*) \geq \tau\big\},$$

where b^* is a trusted reference box. More sophisticated models directly estimate geometric annotation noise, respecting equivalent OBB parameterizations.

8. Further Questions

1. How can unknown p_i values be estimated without perfect reference labels?
2. How should annotation budgets be allocated across annotators with different costs and accuracies?
3. Under what conditions is weighted Bayesian voting superior to ordinary majority voting?
4. Can sequential annotation stop when the posterior uncertainty becomes sufficiently small?
5. How can noisy OBBs be aggregated while respecting geometry and rotational symmetry?

9. References and Further Reading

- Condorcet Jury Theorem — reliability amplification under independent judgments.
- Poisson Binomial Distribution — independent non-identically distributed Bernoulli sums.
- Hoeffding's Inequality — concentration bounds for independent bounded variables.
- Dawid, A. P., and Skene, A. M. (1979). "Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm." *Journal of the Royal Statistical Society, Series C.*

- Bayesian Decision Theory; Crowdsourcing; Active Learning.

Key takeaway. Identical independent annotators lead to a binomial majority-vote model. Unequal independent accuracies lead to a Poisson binomial model. Known annotator reliabilities motivate Bayesian weighted voting. Correlated, item-dependent errors require richer models. More labels help only when they contribute sufficiently reliable and nonredundant information.